

Route Right: A Cost Model and Allocation Engine for Multi-Venue Crypto Execution

Engine specification, v0.1

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Abstract

We formalize the per-period (trailing 30-day) cost a trading firm incurs when its order flow is split across crypto venues with *status-based tiered* fee schedules, native-token discounts, conditional market-maker programs, and finite liquidity. The central structural fact is that explicit fees are non-convex in volume (tier rates step *down* at thresholds, rewarding concentration) while market impact is convex (penalizing it); the optimal route balances the two. We give the cost function, prove the relevant monotonicities, and reduce the allocation problem to a mixed-integer linear program solved exactly at the scale of interest.

1 Setup and notation

Let $\mathcal{V}$ index venues. Over one rolling 30-day window a firm must execute total notional $V > 0$ in N executions, so the average trade size is $S = V/N$. A fraction $\varphi \in [0, 1]$ of notional is executed as *taker* (crossing the spread); the complement $1 - \varphi$ is *maker* (passive). The decision is an allocation

$$x = (x_v)_{v \in \mathcal{V}}, \quad x_v \geq 0, \quad \sum_v x_v = V,$$

where x_v is notional routed to venue v . We assume the firm preserves its order-size distribution across venues, so the executions placed on v number $n_v = N x_v/V$ and retain average size S .

Rates are stored as fractions of notional; $0.0010 = 0.10\% = 10$ bps. A negative maker rate denotes a rebate.

2 Explicit fees

2.1 Tiers as a status

Each venue has an ascending tier ladder with thresholds $0 = \Theta_{v,0} < \Theta_{v,1} < \dots < \Theta_{v,K_v}$ and per-tier maker/taker rates $(m_{v,k}, t_{v,k})$. Crucially, the tier is a *status*: the rate of the attained tier applies to the firm's *entire* volume on v , not marginally. The attained tier index is

$$k_v(q_v) = \max\{k : q_v \geq \Theta_{v,k}\},$$

where the qualifying metric q_v depends on the venue's rule: volume ($q_v = x_v$), assets ($q_v = A_v$), **either** ($q_v = \max(x_v, A_v)$), or **both** ($q_v = \min(x_v, A_v)$), e.g. a venue requiring both volume and a token balance).

2.2 Effective rates

Two adjustments turn posted rates into expected effective rates.

Native-token discount. If the firm pays fees in v 's token, a discount δ_v multiplies *positive* fees only (rebates are unaffected), at the cost of a carry c_v^{tok} (bps on routed notional) for holding the token. The effective taker rate is $\tilde{t}_{v,k} = t_{v,k} (1 - \delta_v \mathbf{1}[\text{pay in token}])$ when $t_{v,k} > 0$.

Expected maker rate. A passive order earns its (possibly negative) rate only if it *fills*. With fill probability p_v , adverse-selection loading a_v , and a chase cost $\approx$ a half-spread h_v on the unfilled remainder,

$$\tilde{m}_{v,k} = p_v (m_{v,k}^\delta + a_v) + (1 - p_v) h_v, \quad (1)$$

where $m_{v,k}^\delta$ is the token-adjusted maker rate. Equation (1) is the correction that prevents the engine from treating a posted rebate as guaranteed income.

Market-maker program (conditional). If the firm is eligible and $\Theta_{v,k} \geq \Theta_v^{\text{mm}}$, an enhanced pair $(m_v^{\text{mm}}, t_v^{\text{mm}})$ is available in exchange for an obligation cost o_v (bps on maker notional). The engine adopts the program on v iff it lowers the blended rate.

The blended explicit rate at tier k is

$$r_{v,k} = \varphi \tilde{t}_{v,k} + (1 - \varphi) \tilde{m}_{v,k} + o_v \mathbf{1}[\text{mm}] + c_v^{\text{tok}} \mathbf{1}[\text{token}], \quad (2)$$

and the explicit fee on v is $r_{v,k_v} x_v$, a *piecewise-linear, non-convex* function of x_v (it drops discontinuously in slope at each $\Theta_{v,k}$).

[Tier monotonicity] For fixed φ and parameters, $r_{v,k}$ is non-increasing in k . Consequently the blended rate is a non-increasing step function of the qualifying metric, and average cost per dollar weakly decreases with volume on a venue. This concavity is the economic incentive to concentrate flow.

3 Implicit cost: spread and impact

Spread. Takers cross the spread each trade: $\text{spread}_v(x_v) = \varphi x_v h_v$, linear in x_v .

Participation impact (convex). Absorbing a share of a venue's window volume ADV_v moves price by a square-root law, giving an aggregate cost

$$I_v^{\text{part}}(x_v) = \kappa \sigma x_v \sqrt{\frac{x_v}{\text{ADV}_v}} = \kappa \sigma \text{ADV}_v^{-1/2} x_v^{3/2}, \quad (3)$$

convex and increasing. This is the counterweight to tier concavity: it makes spreading worthwhile and yields an interior optimum.

Size impact (trade-count effect). Holding notional fixed, fewer (larger) orders dig deeper into the instantaneous book. With instantaneous depth $d_v = \beta \text{ADV}_v$,

$$I_v^{\text{size}}(x_v) = \kappa_S \sigma x_v \sqrt{\frac{S}{d_v}}, \quad S = \frac{V}{N}, \quad (4)$$

linear in x_v but scaling as $\sqrt{S} = \sqrt{V/N}$, so for fixed V the cost *rises as N falls*. Equations (3)–(4) are how the engine honors the notional-versus-trade-count distinction: V sets tiers, S sets per-order impact. Optional flat per-order fees add $n_v \phi_v$ (typically 0 for crypto spot).

4 The allocation problem

Total expected monthly cost is

$$C(x) = \sum_v \left[r_{v,k_v(x_v)} x_v + \varphi h_v x_v + I_v^{\text{part}}(x_v) + I_v^{\text{size}}(x_v) + n_v \phi_v \right]. \quad (5)$$

We minimize C over $\sum_v x_v = V$, the counterparty cap $x_v \leq \rho V$ (share $\rho \in (0, 1]$), access masks $x_v = 0$ for venues the firm’s region cannot reach, and optional semi-continuous minimum lots. Because the first term is non-convex, this is not a convex program.

4.1 MILP formulation

We disaggregate each x_v over its tier brackets with binaries $z_{v,k} \in \{0, 1\}$ and segment variables $y_{v,k} \geq 0$:

$$\sum_k z_{v,k} = 1, \quad x_v = \sum_k y_{v,k}, \quad (6)$$

$$\Theta_{v,k} z_{v,k} \leq y_{v,k} \leq U_{v,k} z_{v,k}, \quad U_{v,k} = \min(\Theta_{v,k+1}, \rho V), \quad (7)$$

so exactly one bracket is active and its rate applies to the whole volume; the explicit term linearizes to $\sum_k r_{v,k} y_{v,k}$. The convex impact (3) is represented by a piecewise-linear outer description with increasing slopes $s_{v,j}$ on segments $\delta_{v,j} \in [0, w_{v,j}]$, $x_v = \sum_j \delta_{v,j}$; convexity makes the LP fill low-slope segments first, so no extra binaries are needed and the approximation is exact at the breakpoints. The remaining terms are linear in x_v . The result is a MILP with $\sum_v K_v$ binaries.

[Scale and exactness] At the scale of interest ($|\mathcal{V}| \sim 6\text{--}15$ venues, $K_v \sim 6\text{--}10$ tiers) the MILP has on the order of 10^2 binaries and CBC solves it to proven optimality in milliseconds. Any single-venue allocation is feasible, so the optimum is provably no worse than the naive “best single venue” baseline; the engine’s edge is exactly the gap.

5 Rolling-tier sprint

Tiers track *trailing* 30-day volume, so mid-window a firm can push extra volume now to re-rate remaining flow. With current trailing volume V_v^{now} at day τ of a 30-day window, run-rate $\mu_v = V_v^{\text{now}}/\tau$, and remaining flow $R_v = \mu_v(30 - \tau)$, crossing the next threshold Θ' (rate $r \rightarrow r'$) is worthwhile iff

$$\underbrace{(r - r') R_v}_{\text{saving on remaining flow}} > \underbrace{r (\Theta' - V_v^{\text{now}})}_{\text{sprint cost}}. \quad (8)$$

The engine reports the breakeven and net benefit per venue.

6 What is data versus structure

The *structure*—tier non-convexity, convex impact, the MILP, the sprint breakeven—is the engine’s defensible content. The following are **parameters to be calibrated** from live feeds and the firm’s own fills, and the engine’s accuracy is bounded by their quality:

- tier thresholds and rates (live per venue; top tiers and MM-program terms are often negotiated/NDA);

- half-spreads h_v , window liquidity ADV_v , depth fraction β , and impact coefficients κ, κ_S (from order-book and TCA data);
- maker fill probabilities p_v and adverse-selection loadings a_v (from the firm's realized passive fills).

Shipping the engine with indicative defaults is fine for illustration; shipping it against a customer's real fills is what makes a route trustworthy.